

AKILA KARUNANAYAKE

Department of Computer Engineering, University of Peradeniya, Sri Lanka 20400

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Interests

Robotics(perception, mapping)

Computer Vision

Edge Computing

Education

University of Peradeniya

Nov. 2018 – Sep. 2024

First-Class Honors in B.Sc. Engineering(Hons.) Computer Engineering

GPA 3.85/4.00

Research Experience

Human Skeleton based Action Recognition

Oct 2024 – July 2025

University of Peradeniya, Sri Lanka

- Using pose estimation heatmaps with attention to accurately predict group activities
- Supervision: Prof. Roshan Godaliyadda

Uncertainty-guided Knowledge Distillation for Stereo Matching

Dec 2023 – Sep 2024

HESL, Nanyang Technological University, Singapore

- Construction of a lightweight and accurate stereo matching deep learning network
- Supervision: Prof. Lam Siew Kei and Dr. Wu Meiqing.
- Implemented and tested the knowledge distillation framework, carried out experiments to reach optimum results and, paper writing. Paper is under review.

Configurable Neuromorphic NoC Architecture for Spiking Neural Networks

May 2023 – Dec 2023

University of Peradeniya, Sri Lanka

- Design and implementation of a RISC-V-based neuromorphic hardware on FPGA for Spiking Neural Networks.
- Supervision: Prof. Roshan Ragel and Dr. Isuru Nawinne.
- Implemented a RISC-V based Network on Chip architecture to support SNNs and tested implementation using a FPGA
- Github : <https://github.com/cepdnaclk/e17-4yp-Neuromorphic-NoC-Architecture-for-SNNs>

Low Cost LIDAR Global Localization

April 2023 – September 2023

*Robotics and Autonomous Systems, I2R, A*STAR, Singapore*

- Researched into low-cost LIDAR global localization of mobile robots.
- Research done as part of A*STAR SIPGA Award. Supervision: Dr. Lawrence Chen and Dr. Saurab Verma.
- Tested novel implementations to match observed key points to recorded key points as 2D images, evaluated performance between graph and image matching, improved OpenCV functions to perform on sparse data, and parallelized to run on GPU using OpenCL

Psuedo RGBD ORBSLAM2

Dec 2022 – May 2023

HESL, Nanyang Technological University, Singapore

- Code implementation of Pseudo RGB-D for Self-Improving Monocular SLAM and Depth Prediction by L.Tiwari et al.
- Supervision: Prof. Lam Siew Kei and Dr. Wu Meiqing.
- Implemented the self-improving framework using Python and C++ and tested the implementation to verify they are inline with the published results
- Github : <https://github.com/Akilax0/Pseudo-RGB-D-for-Self-Improving-Monocular-SLAM-and-Depth-Prediction>

Work Experience

Department of Computer Engineering

March 2020 – July 2025

Volunteer Developer, Maintainer and Instructor (Teaching Assistant)

- Development and maintenance of the following department sites.
 - * <https://projects.ce.pdn.ac.lk/ongoing-projects/>
- Project coordinator for 40+ undergraduates working on different development projects.
- Setup and Maintenance of servers at the Department.
- Casual instructor for Computer Architecture (CO224) and Computer Systems Engineering (CO326).
- Instructor for Image Processing (CO543), Data Structures and Algorithms (CO322), Advanced Communication Networks (CO513), Computer Communication Networks (CO323) .

STERNX (Startup) | <https://www.sternxengineering.com/>

May 2020 – Jan 2022

Junior Software Engineer

- Developed front end for the company depicting the services and blog posts of the employees.
- Utilized Javascript frameworks, HTML, CSS to allow updates on external sites to be displayed on the relevant site .

Projects

Autonomous Landmine Detector <i>C++, Python, AWS, Selenium</i>	Jun 2022
- Developed an autonomous bot controlled by an ESP32 to scan a given area for landmines using electromagnetic methods and display results on a webapp. - Created a back-end using AWS services to store parameters used in each turn and its results. - Technologies: ESPIDF, MQTT, I2C, SPI . - Github : https://github.com/cepdnaclk/e17-3yp-Landmine-Detector	
Smart Building <i>Automation, IoT</i>	Oct 2022
- Project lead for a group of 60 undergraduates. - Design and prototype implementation of the system. - Technologies: MQTT, NodeRED, Docker, Arduino. - Github : https://github.com/cepdnaclk/e17-co326-Smart-Building	
Analysis Tool for Industrial Images <i>OpenCV , Automation</i>	Feb 2022
- Created a tool to analyze the performance of an image processing algorithm used to detect deformities in an industrial molding machine. - Dashboard and API were created to visualize the results. - Technologies: OpenCV, React, ExpressJS, WebSocket. - Github : https://github.com/cepdnaclk/e17-co328-Analysis-Tool-for-Industrial-Images	
Compiler for Cool Language <i>COOL, C++</i>	Feb 2022
- The combination of a lexer, parser, semantic analyser, and code generator that can be used to compile programs written in Cool programming language. - Github : https://github.com/Akilax0/assignments	
Vehicle Number Plate Analyzer <i>Image Processing, OCR</i>	Feb 2022
- Created Tool to analyze CCTV captured images and recognize number plates of vehicles. - Classical image processing techniques were used to remove noise and scale the raw images such as super-resolution, histogram analysis, and Fourier domain analysis. - Optical character recognition used to extract information from the resulting images. - Report: https://drive.google.com/file/d/14ejy8Z_6T3mxUF3Oj9dBymhuGgTtWvGL/view?usp=sharing	
8-bit processor <i>Verilog, ARM assembly</i>	October 2020
- Designed 8-bit ALU with a register file for memory using Verilog. - Simulated processor behavior using Icarus Verilog and input and output signals were observed using GTKWave. - Tested behavior using ARM assembly code. - Github : https://github.com/Akilax0/FPGA_CO503/tree/main/CO224	
Image Processing techniques to detect damaged fruit <i>Python, OpenCV</i>	November 2019
- Image Filtering with OpenCV was used to create an algorithm to detect the deformities of fruit . - Created application using python to continuously monitor given set of images .	

Competitions

1st place at ACES Coders (of 120+ teams)	2022
<i>12 hour competitive programming competition for university undergraduates in Sri Lanka.</i>	
1st place at Code Squad v3.0 (150+ teams)	2022
<i>6 hour competitive programming competition for university undergraduates in Sri Lanka.</i>	
1st and 2nd Runner up of MoraXtreme 6.0 and 7.0 respectively (of 200+ teams)	Oct.2021/22
<i>12 hour competitive programming competition for university undergraduates in Sri Lanka.</i>	
185th and 142nd world rank of IEEE Xtreme 15.0 and 16.0 respectively	Oct.2021/22
<i>24 hour competitive programming competition for university undergraduates worldwide.(out of 6000+ teams)</i>	
5th place at IESL UIY	2021
<i>Undergraduate innovator of the Year competition organized by IESL for undergraduates of Sri Lanka</i>	
Jaffna Coders Competitive Programming Competition	2019
<i>Entered the Final 20 teams out of 100+ teams</i>	
Top 20 country rank of Google Code Jam, ACES Coders	2019-2022

Technical Skills

Languages	C, C++, Verilog HDL, Python, Java, HTML/CSS, JavaScript
Developer Tools	ESP-IDF, Quartus, GTKWave, AWS, Android Studio
Technologies/Frameworks	Linux, Git, Pytorch, OpenCV, Tensorflow, Keras, Jekyll

Extracurricular

Teaching Git & Github Fundamentals with Hackers' Club for all undergraduates	2021
<i>Workshop to introduce basic developer skills</i>	
Slides: https://drive.google.com/drive/folders/18zGvksfkHTUNqcctLs4e_bIR5jXdUOgL?usp=sharing	
Member of the Web Consultation team of University of Peradeniya	2021- 2022
<i>Group focused on improving university's digital presence</i>	
Swarm Robotics group	2021- 2025
<i>Documentation and project supervision</i>	